

Fig. 8.18

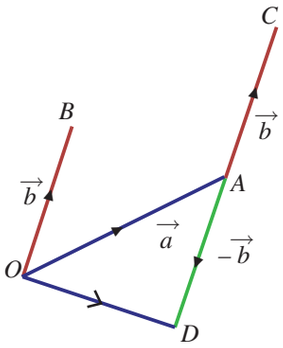


Fig. 8.19

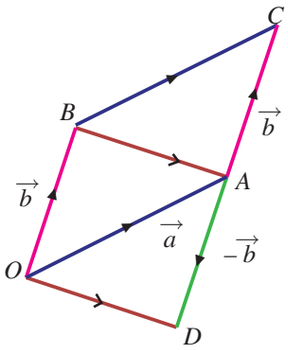


Fig. 8.20

Thus, **if** $\vec{a}$ and $\vec{b}$ **represent two adjacent sides of a parallelogram then the diagonals represent** $\vec{a}+\vec{b}$ **and** $\vec{a}-\vec{b}$ **.**

8.4.3 Scalar multiplication of a vector

Now let us see how to multiply a vector by a scalar.

Let $\vec{a}$ be a vector and m be a scalar. Then the vector $m\vec{a}$ is called the **scalar multiple of a vector** $\vec{a}$ by the scalar m.

Let us note that when m is zero, the magnitude of $m\vec{a}$ becomes 0 and hence $m\vec{a}$ becomes the zero vector. If m is positive, then both $\vec{a}$ and $m\vec{a}$ have the same direction and when m is negative, then $\vec{a}$ and $m\vec{a}$ have opposite directions. Thus $\vec{a}$ and $m\vec{a}$ are like vectors if m is positive and unlike vectors if m is negative. The magnitude of $m\vec{a}$ is $|m\vec{a}| = |m||\vec{a}|$.

Definition 8.13

Two vectors $\vec{a}$ and $\vec{b}$ are said to be **parallel** if $\vec{a} = \lambda\vec{b}$, where λ is a scalar. If $\lambda > 0$, they are in the same direction. If $\lambda < 0$ then they are in the opposite direction to each other.

8.4.4 Some properties and results

For any two vectors $\vec{a}$ and $\vec{b}$ and scalars m and n, we have

(i) $m(n\vec{a}) = mn(\vec{a}) = n(m\vec{a})$

(ii) $(m+n)\vec{a} = m\vec{a} + n\vec{a}$

(iii) $m(\vec{a}+\vec{b}) = m\vec{a} + m\vec{b}$

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Result 8.2

Vector addition is associative.

For any three vectors $\vec{a}, \vec{b}$ and $\vec{c}$,

$$(\vec{a}+\vec{b})+\vec{c} = \vec{a}+(\vec{b}+\vec{c}).$$

Result 8.3

For any vector $\vec{a}$, $\vec{a}+\vec{0} = \vec{0}+\vec{a} = \vec{a}$.

Result 8.4

For any vector $\vec{a}$, $\vec{a}+(-\vec{a}) = (-\vec{a})+\vec{a} = \vec{0}$.

This result states that the additive inverse exists for every vector.

Result 8.5

Vector addition is commutative.

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